Logic Programming

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Lecture #12

Graph isomorphism

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Agenda

- **Graphs isomorphism**
  - with metaprogramming
  - with nonmonotonic reasoning
  - various implementations (compare & contrast analysis)

Graphs isomorphism problem statement

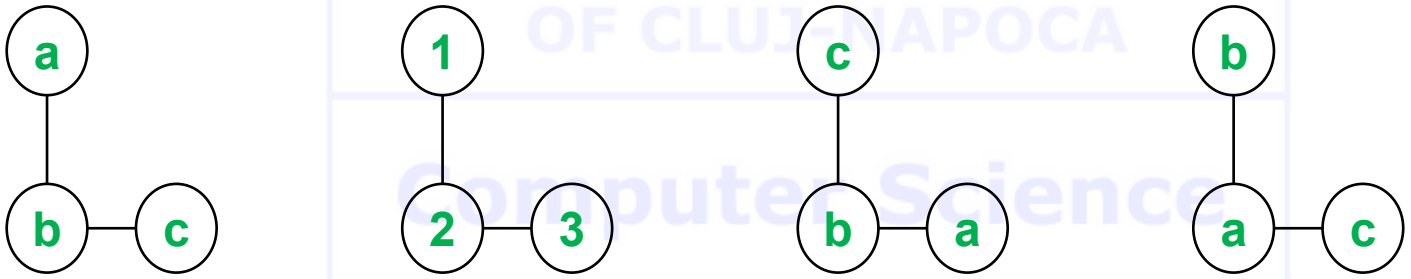
- Let us:
  - G1=(V1,E1)
  - G2=(V2,E2)
  - G1 iso G2 iff
    i) |V1|=|V2|
    ii) $\exists\ \varphi$: V1-> V2 a bijection so that
    iii) $\forall$ x, y $\in$V1 [(x, y) $\in$ E1 $\Leftrightarrow$ (φ(x), φ(y)) $\in$ E2)
- In simpler words:
  - Graphs are essentially "the same" in structure, differing only by the names or arrangement of their vertices.
  - φ matches up vertices one-to-one while preserving adjacency
- Ex: Are the 2 graphs isomorphs?

```
graph1([
        n(a,[b,c,d]),
        n(b,[a,c]),
        n(c,[a,b,d]),
        n(d,[a,c])
]).
```

```
graph2([
        n(1,[2,4]),
        n(2,[1,3,4]),
        n(3,[2,4]),
        n(4,[1,2,3])
]).
```

Graphs isomorphism
Simple example

- Consider these two graphs:
  - **Graph A** has vertices labelled {a,b,c} with edges (a,b),(b,c).
  - **Graph B** has vertices labelled {1,2,3} with edges (1,2),(2,3).



Graphs isomorphism approach (v1)

```
?- graph1(G1), graph2(G2), iso_graph1(G1,G2).

iso_graph1(L1,L2):- eq_perm(L1,L2,eq_neighb).

eq_perm([H1|T1],L2,EQ):-      % is the perm predicate with an equivalence
       delete(H2,L2,T2),      % nondeterministic behavior. Implemented how?
       P=..[EQ,H1,H2],        % which is created here (metapredicate)
       call(P),               % and called here to execute
       eq_perm(T1,T2,EQ).
eq_perm([],[],_).             % succeeds iff sets have the same cardinality
% an immediate improvement would check cardinalities beforehand

eq_neighb(n(N1,L1),n(N2,L2)):-   % 2 neighb pairs are equivalent
       eq_node(N1,N2),           % if the nodes are equivalent
       eq_perm(L1,L2,eq_node).   % and the neighb lists are equivalent

delete(H,[H|T],T).               % what if we add a cut here?
delete(X,[H|T],[H|R]):-
       delete(X,T,R).            % what if we add the [] clause?
```

Nodes equivalence
(implements φ function)
V1 – default logic with side effects

- p a dynamic predicate, used for the nomonotonic reasoning
- implements φ function (to form p as akb)

```
eq_node(N1,N2):-    % if nodes N1 and N2 already form a pair in the akb
      p(N1,N2).     % the evaluation continues
eq_node(N1,_):-     % if N1 forms a pair with some OTHER node
      p(N1,_),!,    % we get an inconsistency, so should NOT allow
      fail.         % (N1,N2) form a pair, so, fail to backtrack
eq_node(_,N2):-     % symmetric on N2
      p(_,N2),!,    % not an injective function
      fail.         % How/where bijection is insured?
eq_node(N1,N2):-    % if you reach here, there is no inconsistency
      asserta(p(N1,N2)).          % hence pair N1, N2 is legitimate.
eq_node(N1,N2):-                  % if at a later moment,
      retract(p(N1,N2)),!,        % although pair N1, N2 is legitimate,
                                  % the reasoning cannot conclude,
                                  % remove it from the akb, and
      fail.                       % fail to backtrack and
                                  % resume execution WITHOUT the pair in the akb
```

Graphs isomorphism approach (v1)

```
?- graph1(G1), graph2(G2), iso_graph1(G1,G2).

iso_graph1(L1,L2):- eq_perm(L1,L2,eq_neighb).

eq_perm([H1|T1],L2,EQ):-
        delete(H2,L2,T2),
        P=..[EQ,H1,H2],
        call(P),
        eq_perm(T1,T2,EQ).
eq_perm([],[],_).

eq_neighb(n(N1,L1),n(N2,L2)):-
        eq_node(N1,N2),
        eq_perm(L1,L2,eq_node).

delete(H,[H|T],T).
delete(X,[H|T],[H|R]):- delete(X,T,R).

eq_node(N1,N2):- p(N1,N2).
eq_node(N1,_):- p(N1,_), !, fail.
eq_node(_,N2):- p(_,N2), !, fail.
eq_node(N1,N2):- asserta(p(N1,N2)).
eq_node(N1,N2):- retract(p(N1,N2)), !, fail.
```

Nodes equivalence
(implements φ function)
V2 – with arguments (lists)

```
?- graph1(G1), graph2(G2), iso_graph2(G1,G2).
% instead of the akb p in v1, here we store the pairs in the arg list
% arg 4 = partial list, empty initially (arg to model the akb)
% arg 5 = final list; a copy of the partial list at the end of the execution
%                  (arg with the status of the akb at the end; the bijection)
iso_graph2(L1,L2):- eq_perm(L1,L2,eq_neighb,[],Lout).

eq_perm([H1|T1],L2,EQ,LI,LO):-      % same as in v1
       delete(H2,L2,T2),
       P=..[EQ,H1,H2,LI,Lint],      % same but with args
       call(P),
       eq_perm(T1,T2,EQ,Lint,LO).
eq_perm([],[],_,L,L).

eq_neighb(n(N1,L1),n(N2,L2),LI,LO):-
       eq_node(N1,N2,LI,Lint),
       eq_perm(L1,L2,eq_node,Lint,LO).
```

Nodes equivalence
(implements φ function)
V2 – with arguments (lists) – contd.

- p is a pair of nodes in the list argument
- list evolves nomonotonic (during reasoning) to allow for pairs addition/removal
- models φ function

```
eq_node(N1,N2,LI,LI):-
        member(p(N1,N2),LI), !.
eq_node(N1,N2,LI,[p(N1,N2)|LI]):-
        not(member(p(N1,_),LI)),
        not(member(p(_,N2),LI)).
```

- These 2 clauses do the same as those 5 in v1

Nodes equivalence
(φ function v2 VS v1)

```
eq_node(N1,N2,LI,LI):-
        member(p(N1,N2),LI),!.

eq_node(N1,N2,LI,[p(N1,N2)|LI]):-

        not(member(p(N1,_),LI)),

        not(member(p(_,N2),LI)).
```

```
eq_node(N1,N2):-
        p(N1,N2).
eq_node(N1,_):-
        p(N1,_),!,
        fail.
eq_node(_,N2):-
        p(_,N2),!,
        fail.
eq_node(N1,N2):-
        asserta(p(N1,N2)).
eq_node(N1,N2):-
        retract(p(N1,N2)),!,
        fail.
```

Nodes equivalence
(implements φ function)
V3 – with arguments (incomplete lists)

- Same as v2 but lists are incomplete
- So, it must adjust the behavior of the search predicate to handle appropriately the "not found" case
- Just the equivalence changes
  - (and the `eq_perm` predicate needs just 1 list arg):

```
eq_node(N1,N2,[p(N1,N2)|_]):-!.
eq_node(N1,_,[p(N1,_)|_]):- !,
      fail.
eq_node(_,N2,[p(_,N2)|_]):- !,
      fail.
eq_node(N1,N2,[_|T]):-
      eq_node(N1,N2,T).
```

- Compare v3 with v1. Why is clause 1 (from v1) missing (in v3)?
- Compare v3 with v2.

Nodes equivalence
(φ function v3 VS v1)

```
eq_node(N1,N2,[p(N1,N2)|_]):-!.
eq_node(N1,_,[p(N1,_)|_]):-
        !,
        fail.
eq_node(_,N2,[p(_,N2)|_]):-
        !,
        fail.
eq_node(N1,N2,[_|T]):-
        eq_node(N1,N2,T).
```

```
eq_node(N1,N2):-
        p(N1,N2).
eq_node(N1,_):-
        p(N1,_),!,
        fail.
eq_node(_,N2):-
        p(_,N2),!,
        fail.
eq_node(N1,N2):-
        asserta(p(N1,N2)).
eq_node(N1,N2):-
        retract(p(N1,N2)),!,
        fail.
```

Nodes equivalence
(φ function v3 VS v2)

```
eq_node(N1,N2,[p(N1,N2)|_]):-!.
eq_node(N1,_,[p(N1,_)|_]):-
        !,
        fail.
eq_node(_,N2,[p(_,N2)|_]):-
        !,
        fail.
eq_node(N1,N2,[_|T]):-
        eq_node(N1,N2,T).
```

```
eq_node(N1,N2,LI,LI):-
        member(p(N1,N2),LI),!.
eq_node(N1,N2,LI,[p(N1,N2)|LI]):-
        not(member(p(N1,_),LI)),

        not(member(p(_,N2),LI)).
```

Nodes equivalence
(implements φ function)
V4 – with arguments (incomplete lists)

- Same as v2 just that lists are incomplete
- So, it must adjust the behavior of the search predicate to handle appropriately the "not found" case
- Just the equivalence changes
  - (and the `eq_perm` predicate needs just 1 list arg):

```
eq_node4(N1,N2,LI):-
       member_il(p(N1,N2),LI), !.
eq_node4(N1,N2,[p(N1,N2)|LI]):-
       not(member_il(p(N1,_),LI)),
       not(member_il(p(_,N2),LI)).

member_il(_, L):-var(L), !, fail.
member_il(X, [X|_]):-!.
member_il(X, [_|T]):-member_il(X, T).
```

- Compare v4 with v2.

Nodes equivalence
(φ function v4 VS v2)

```
eq_node(N1,N2,LI):-
        member_il(p(N1,N2),LI),!.

eq_node(N1,N2,[p(N1,N2)|LI]):-

        not(member_il(p(N1,_),LI)),

        not(member_il(p(_,N2),LI)).
```

```
eq_node(N1,N2,LI,LI):-
        member(p(N1,N2),LI),!.

eq_node(N1,N2,LI,[p(N1,N2)|LI]):-

        not(member(p(N1,_),LI)),

        not(member(p(_,N2),LI)).
```